S6: Regression, Correlation and Hypothesis Testing

1. Scatter Diagrams and Correlation

A **scatter diagram** plots bivariate data (two variables, one on each axis) to reveal the relationship between them. The independent variable is placed on the x-axis; the dependent variable on the y-axis.

Correlation describes the linear relationship between two variables. It can be:

- **Positive** — as x increases, y tends to increase
- **Negative** — as x increases, y tends to decrease
- **Zero (none)** — no linear relationship visible

Strength: *strong* (points close to a line), *moderate*, or *weak* (widely scattered).

Exam tip: Correlation does NOT imply causation. Always state this explicitly if asked. Two variables can be correlated due to a *lurking/confounding variable* or pure coincidence.

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2. Product Moment Correlation Coefficient (PMCC)

PMCC, denoted r, measures the strength and direction of the *linear* relationship between two variables. It always satisfies $-1 \leq r \leq 1$.

$$r = \frac{S_{xy}}{\sqrt{S_{xx} \cdot S_{yy}}}$$

where the summary statistics are:

$$S_{xy} = \sum xy - \frac{\sum x \sum y}{n}, \quad S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}, \quad S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}$$

Value of r	Interpretation
$r = 1$	Perfect positive linear correlation
$0 < r < 1$	Positive linear correlation (closer to 1 = stronger)
$r = 0$	No linear correlation
$-1 < r < 0$	Negative linear correlation (closer to -1 = stronger)
$r = -1$	Perfect negative linear correlation

Exam tip: r measures only *linear* correlation. A value near 0 does not mean no relationship — there could be a strong non-linear relationship.

Worked Example — Calculating PMCC

Example: For $n = 5$ data pairs, $\sum x = 20$, $\sum y = 30$, $\sum xy = 135$, $\sum x^2 = 90$, $\sum y^2 = 200$. Find r.

$$S_{xy} = 135 - \frac{20 \times 30}{5} = 135 - 120 = 15$$
$$S_{xx} = 90 - \frac{20^2}{5} = 90 - 80 = 10$$
$$S_{yy} = 200 - \frac{30^2}{5} = 200 - 180 = 20$$
$$r = \frac{15}{\sqrt{10 \times 20}} = \frac{15}{\sqrt{200}} = \frac{15}{14.14...} \approx 0.671$$

This indicates moderate positive linear correlation.

3. Linear Regression

The **regression line of y on x** (also called the least squares regression line) minimises the sum of squared *vertical* residuals. Its equation is:

$$\hat{y} = a + bx$$

where the gradient and intercept are calculated as:

$$b = \frac{S_{xy}}{S_{xx}}, \qquad a = \bar{y} - b\bar{x}$$

Here $\bar{x} = \dfrac{\sum x}{n}$ and $\bar{y} = \dfrac{\sum y}{n}$ are the sample means. The regression line always passes through $(\bar{x}, \bar{y})$.

Residual for a data point (x_i, y_i): the difference $e_i = y_i - \hat{y}_i$. The regression line makes $\sum e_i^2$ as small as possible.

Using the Regression Line

- Use the **y on x** line to *predict y from a given x value only*.
- **Interpolation** — predicting within the range of the data. Generally reliable.
- **Extrapolation** — predicting outside the range of the data. Unreliable; the model may not hold beyond the data.

Exam tip: Never use the y on x regression line to predict x from y. If asked to predict x, you need the x on y regression line (which is NOT the rearrangement of y on x).

Example: From the data above, find the regression line of y on x.

$\bar{x} = \dfrac{20}{5} = 4$, \quad $\bar{y} = \dfrac{30}{5} = 6$

$b = \frac{15}{10} = 1.5$
$a = 6 - 1.5 \times 4 = 6 - 6 = 0$
So the regression line is $\hat{y} = 1.5x$.

Predict y when x = 3 (interpolation): $\hat{y} = 1.5 \times 3 = 4.5$.

4. Coding Data for Regression

When data is **coded** (transformed) to simplify calculations, the regression equation must be decoded to express y in terms of x (original variables).

Common coding: $p = \frac{x-a}{b}$ and $q = \frac{y-c}{d}$. If the regression of q on p is $q = \alpha + \beta p$, then substitute back:

$$\frac{y-c}{d} = \alpha + \beta \cdot \frac{x-a}{b}$$

Then rearrange for y in terms of x.

Exam tip: The PMCC r is unaffected by linear coding (it is dimensionless). You get the same value of r whether you use raw or coded data.

5. Hypothesis Testing for ρ (Population PMCC)

The **population PMCC** ρ (rho) measures the true underlying linear correlation. The sample PMCC r is an estimate of ρ. A hypothesis test determines whether observed correlation is statistically significant.

Setting Up the Test

- H_0: $\rho = 0$ (no linear correlation in the population)
- H_1: $\rho \neq 0$ (two-tailed, testing for any correlation)

- H_1: $\rho > 0$ (one-tailed, testing for positive correlation)
- H_1: $\rho < 0$ (one-tailed, testing for negative correlation)

Procedure

1. State H_0 and H_1 clearly, including the significance level α.
2. Calculate the sample PMCC r from the data.
3. Look up the critical value r_c from the Edexcel critical values table (using n and α).
4. Compare: if $|r| > r_c$ (two-tailed) or $r > r_c$ (one-tailed positive), reject H_0.
5. State conclusion in context.

Example: A researcher collects $n = 10$ pairs of data and calculates $r = 0.65$. Test, at the 5% significance level, whether there is evidence of positive correlation.

$H_0 : \rho = 0$, \quad $H_1 : \rho > 0$ (one-tailed, 5%)
Critical value from table for $n = 10$, one-tailed 5%: $r_c = 0.5494$
Since $0.65 > 0.5494$, we **reject** H_0.
There is sufficient evidence at the 5% significance level to conclude that there is positive linear correlation between the two variables.

Exam tip: Always look up the correct critical value for your n and significance level. For a two-tailed test at 5%, use the 2.5% column. State the conclusion in the context of the problem — say what the variables are.

6. Key Formulae Summary

Formula	Description
$S_{xy} = \sum xy - \frac{\sum x \sum y}{n}$	Sum of products (coded)
$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$	Sum of squares for x

$S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}$	Sum of squares for y
$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$	Sample PMCC
$b = \frac{S_{xy}}{S_{xx}}$	Gradient of regression line
$a = \bar{y} - b\bar{x}$	Intercept of regression line